9.1.3 Open source CP tooling

Let us first have a look at open source tooling in the continuous profiling space. This is a domain that exploded around 2020, and we see increasing interest in it to the extent that OpenTelemetry kicked off an activity in September 2022 (see also OTEP 139, http://mng.bz/JgjV) to explore how to support profiles as a signal type. In alphabetical order, we will now have a look at popular OSS tooling for continuous profiling.

PARCA

Parca (https://www.parca.dev/) is a Prometheus-inspired open source continuous profiling tool that specifically focuses on zero-instrumentation using eBPF, so you don't have to change anything about your application or drop any libraries into your code to make it work. It uses Prometheus service discovery to find profile sources. Parca supports a long list of programming languages (https://www.parca.dev/docs/parca-agent-language-support), including C, C++, Rust, Haskell, Java, Python, .NET, and WebAssembly.

You can install it and try it out yourself or use the zero-install online environment at demo.parca.dev (https://demo.parca.dev/) to give it a try. Figure 9.4 shows a CPU profile of a Kubernetes application in Parca, and at the end of this section, we will use Parca in an end-to-end walk-through.

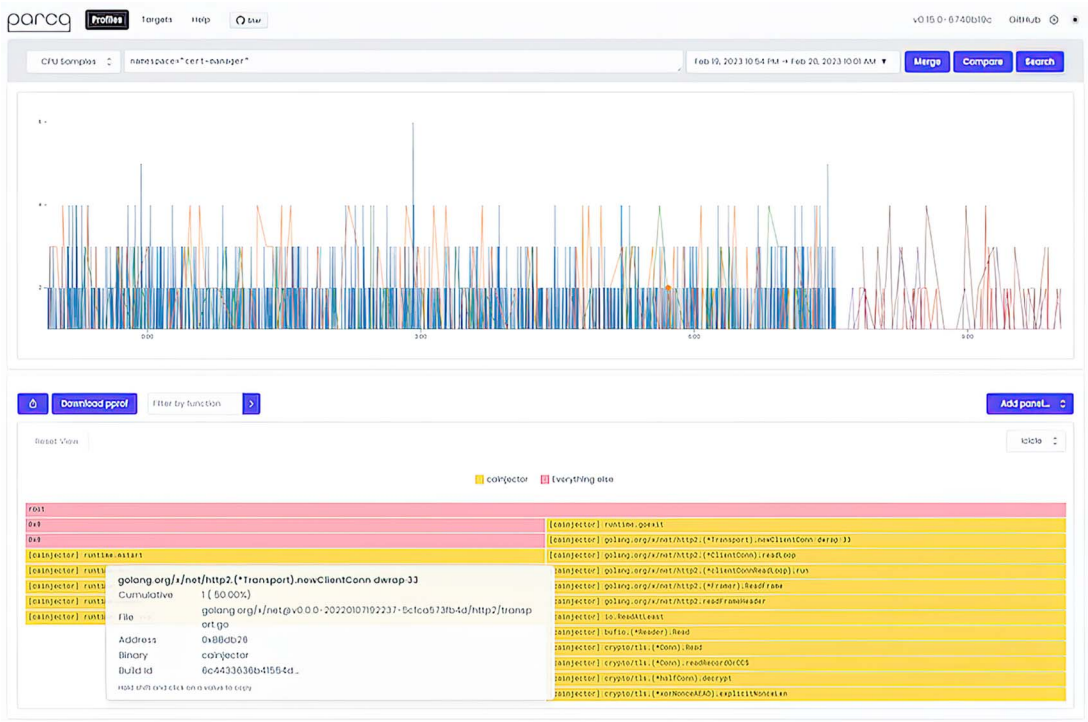


Figure 9.4 Screenshot of the Parca demo environment